

Half letter frequency and balanced borders for one forbidden binary word

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Abstract

Let $B_n(w)$ be the binary words of length n avoiding a fixed nonempty word w . We prove that the mean proportion of ones in $B_n(w)$ tends to $1/2$ if and only if every nonempty border of w , including w itself, contains equally many zeros and ones. This establishes the limiting-frequency part of Conjecture 6.1 of Bóna, Maga and Richey. The known sufficiency direction is rederived from a first-occurrence generating function. For necessity, an Abelian argument forces a border-imbalance polynomial to vanish at the exponential growth rate. A finite-sum bound, controlled by the longest unbalanced border and the exact growth equation, excludes that vanishing even when border imbalances have mixed signs. The argument does not assume that the generating function has a simple pole.

1 Statement and context

For a binary word v , let $|v|_1$ and $|v|_0$ denote its numbers of ones and zeros. Fix $w \in \{0, 1\}^k$, where $k \geq 1$, and define

$$B_n(w) = \{v \in \{0, 1\}^n : w \text{ is not a contiguous factor of } v\}, \quad r_n(w) = \frac{\sum_{v \in B_n(w)} |v|_1}{n|B_n(w)|} \quad (n \geq 1).$$

A nonempty *border* of w is a word that is both a prefix and a suffix of w . The whole word is included. A word is *balanced* if its numbers of zeros and ones agree.

Theorem 1. *For every nonempty finite binary word w ,*

$$\lim_{n \rightarrow \infty} r_n(w) = \frac{1}{2} \iff \text{every nonempty border of } w \text{ is balanced.}$$

When these conditions hold, $r_n(w) = 1/2$ for every $n \geq 1$.

The limiting value is denoted by ρ_w in Bóna, Maga and Richey [1]. Their Proposition 3.5 proves balanced-border sufficiency, in fact through symmetry of the finite distribution, and their Theorem 3.7 treats the case when all nonzero border imbalances have one sign. Their Conjecture 6.1 asks for the converse without that sign restriction, and also states a separate conjecture for a statistic q_w . Theorem 1 proves exactly the ρ_w equivalence. The q_w analogue and the ordering conjecture in [1] are outside the present result.

The weighted border generating function below is the identity in [1, Fact 2.9], derived here by a first-occurrence bijection. The additional step is a dominance bound that allows any number of

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oppositely signed border imbalances. Its finite endpoint term is essential: it can be bounded using the same border polynomial that determines the growth rate. Generating functions, coefficient estimates and exact finite verification also occur in the adjacent enumerative probability work [2]; that reference is methodological context and supplies no forbidden-word theorem used here.

We formulate the left side of Theorem 1 as convergence to $1/2$. Necessity assumes precisely that convergence; sufficiency proves equality at every finite length. Thus the proof requires no general existence theorem for limits at other values, nor any choice of a maximal-entropy measure. A bounded literature search completed on September 7, 2026 found no equivalent resolution in the inspected sources; this observation is not a claim of exhaustive priority.

2 The first-occurrence identities

Let $I \subseteq \{1, \dots, k\}$ be the set of nonempty border lengths. Write $m = |w|_1$, and, for $j \in I$, put

$$m_j = |w[1, j]|_1, \quad d_j = 2m_j - j.$$

Here $w[1, j]$ denotes the prefix of length j , and $k \in I$. Define the polynomials

$$G(x, z) = \sum_{j \in I} x^{k-j} z^{m-m_j}, \quad H(x) = G(x, 1), \quad (1)$$

$$J(x) = \sum_{j \in I} d_j x^{k-j}, \quad P(x, z) = (1 - x(1 + z))G(x, z) + x^k z^m. \quad (2)$$

We abbreviate $P(x, 1)$ by $P(x)$.

Let $U(x, z)$ count the avoiding words, including the empty word, with x marking length and z marking ones. Let $V(x, z)$ count words whose first occurrence of w ends at their final position. Appending a letter to an avoiding word gives

$$(1 - x(1 + z))U = 1 - V, \quad x^k z^m U = VG. \quad (3)$$

For the second identity, append w to an avoiding word of length n . Its first occurrence of w ends at $n + j$ for a unique $1 \leq j \leq k$. Its overlap with the terminal copy consists of j letters, so $j \in I$. The resulting word is this first-occurrence prefix followed by $w[j + 1, k]$; the latter has weight $x^{k-j} z^{m-m_j}$. Conversely, append $w[j + 1, k]$ to a first-occurrence word u , for any $j \in I$. The final k letters form w , and the prefix before this terminal copy has length $|u| - j < |u|$, so it avoids w . The constructions are inverse and preserve both weights.

Eliminating V yields the formal power-series identity

$$U(x, z) = \frac{G(x, z)}{P(x, z)}. \quad (4)$$

The denominator has constant term 1. Set

$$a_n = |B_n(w)|, \quad s_n = \sum_{v \in B_n(w)} (2|v|_1 - n), \quad a_0 = 1, \quad s_0 = 0.$$

Lemma 2. *The ordinary and centered counting series satisfy*

$$A(x) := \sum_{n \geq 0} a_n x^n = \frac{H(x)}{P(x)}, \quad S(x) := \sum_{n \geq 0} s_n x^n = -\frac{x^k J(x)}{P(x)^2}. \quad (5)$$

Proof. Apply $L = 2z\partial_z - x\partial_x$ to (4) and set $z = 1$. Writing $d_k = 2m - k$, direct differentiation gives

$$(LG)(x, 1) = d_k H - J, \quad (L(1 - x(1 + z)))(x, 1) = 0,$$

and hence

$$(LP)(x, 1) = d_k P - (1 - 2x)J.$$

The quotient-rule numerator is

$$(d_k H - J)P - H(d_k P - (1 - 2x)J) = -x^k J,$$

which proves the second identity. The first is (4) at $z = 1$. $\square$

If every border is balanced, $J = 0$, so $s_n = 0$ for every n . Since $s_n/(na_n) = 2r_n(w) - 1$ for $n \geq 1$, this proves sufficiency and the finite equality in Theorem 1.

3 An Abelian necessary condition

Define

$$C(t) = \sum_{j \in I} t^{j-1}, \quad D(t) = \sum_{j \in I} d_j t^{j-1}. \quad (6)$$

Lemma 3. *The limit $\lambda = \lim_{n \rightarrow \infty} a_n^{1/n}$ exists, belongs to $[1, 2)$, and satisfies*

$$(2 - \lambda)C(\lambda) = 1. \quad (7)$$

If $r_n(w) \rightarrow 1/2$, then $D(\lambda) = 0$.

Proof. Cutting an avoiding word into two pieces gives $a_{n+r} \leq a_n a_r$. At least one constant word avoids w , so all a_n are positive. The elementary subadditive argument therefore gives

$$\lambda = \inf_{n \geq 1} a_n^{1/n}, \quad a_n \geq \lambda^n.$$

Indeed, for a fixed $r \geq 1$, write $n = qr + h$, $0 \leq h < r$, and use $\log a_n \leq q \log a_r + \log a_h$; division by n , followed by the infimum over r , proves the asserted limit. Inspecting disjoint blocks of length k gives

$$a_{qk+h} \leq (2^k - 1)^q 2^h \quad (0 \leq h < k), \quad 1 \leq \lambda \leq (2^k - 1)^{1/k} < 2.$$

The radius of convergence of A is $R = 1/\lambda$. For $0 < x < R$,

$$A(x) \geq \frac{1}{1 - \lambda x}, \quad xA'(x) \geq \frac{\lambda x}{(1 - \lambda x)^2}. \quad (8)$$

Both diverge as $x \uparrow R$. As $H(R) > 0$, the identity $PA = H$ forces $P(R) = 0$, while $P(x) = H(x)/A(x) > 0$ for $0 < x < R$.

Now suppose $r_n(w) \rightarrow 1/2$, equivalently $s_n = o(na_n)$. We claim

$$\frac{S(x)}{xA'(x)} \longrightarrow 0 \quad (x \uparrow R). \quad (9)$$

For any $\varepsilon > 0$, choose N such that $|s_n| \leq \varepsilon na_n$ for $n \geq N$. The absolute value of the corresponding tail of S is at most $\varepsilon xA'(x)$. Its finite initial part stays bounded, whereas $xA'(x)$ diverges by (8). Also $|s_n| \leq na_n$ ensures absolute convergence for $x < R$. This proves (9).

Differentiating $A = H/P$ in (5) gives, throughout $(0, R)$,

$$\frac{S(x)}{xA'(x)}(H'(x)P(x) - H(x)P'(x)) = -x^{k-1}J(x). \quad (10)$$

The polynomial factor on the left has a finite limit at R . Thus (9) implies $J(R) = 0$, without dividing by $P'(R)$. Finally, direct substitution shows

$$P(1/t) = t^{-k}((t-2)C(t) + 1), \quad J(1/t) = t^{1-k}D(t).$$

Together with $P(R) = 0$ these give both conclusions. $\square$

Lemma 4. *If $k \geq 3$, then $\lambda \geq \varphi = (1 + \sqrt{5})/2 > 1$.*

Proof. Choose a length-three factor u of w . If u contains 00 or 11, words avoiding that equal-letter pair also avoid w . Their counts obey the Fibonacci recurrence and have exponential growth φ . The remaining possibilities are 010 and 101. For 010, use all concatenations of the prefix code $\{0, 11\}$: every run of ones has even length, so 010 is absent. For 101, use $\{1, 00\}$ and even runs of zeros. In either case unique decoding gives distinct words, and the number f_n of total length n satisfies $f_0 = f_1 = 1$ and $f_n = f_{n-1} + f_{n-2}$. Since these words avoid u and therefore w , we have $a_n \geq f_n$ and the result follows. $\square$

4 Dominance of the longest unbalanced border

Proposition 5. *Assume $k \geq 3$ and some border is unbalanced. Let $s = \max\{j \in I : d_j \neq 0\}$. Then $D(\lambda)$ has the sign of d_s . If $s \geq 3$, more precisely,*

$$\text{sgn}(d_s)D(\lambda) \geq \frac{|d_s|(s-2)}{\lambda-1} > 0. \quad (11)$$

Proof. Complementing every letter of w preserves its border lengths and λ , while reversing all d_j . We may assume $b = d_s \geq 1$. For an earlier border $j < s$ with $d_j < 0$, set $\ell = s - j$. The two prefixes differ by ℓ letters, each contributing at most one to the imbalance difference. Consequently

$$b - d_j \leq \ell, \quad \ell \geq b + 1 \geq 2, \quad \frac{-d_j}{b} \leq \frac{\ell}{b} - 1 \leq \ell - 1. \quad (12)$$

All longer borders have zero imbalance. Discarding earlier positive terms and inserting the bounds for every possible missing negative term therefore gives, for $t > 1$,

$$\frac{D(t)}{bt^{s-1}} \geq 1 - \sum_{\ell=2}^{s-1} (\ell-1)t^{-\ell}. \quad (13)$$

If $s \leq 2$, there are no earlier negative terms by (12), and $D(t) > 0$.

Suppose $s \geq 3$. Multiplication by $t^{s-1}(t-1)^2$, or differentiation of a finite geometric sum, proves the exact identity

$$1 - \sum_{\ell=2}^{s-1} (\ell-1)t^{-\ell} = \frac{(s-1)t - (s-2) - (2-t)t^s}{t^{s-1}(t-1)^2}. \quad (14)$$

Set $t = \lambda$, so $1 < t < 2$ by Lemmas 3 and 4. The whole-word border gives $C(t) \geq t^{k-1}$. Using (7) and $s \leq k$, we obtain

$$(2-t)t^s = \frac{t^s}{C(t)} \leq t^{s-k+1} \leq t. \quad (15)$$

Thus the numerator in (14) is at least $(s-2)(t-1)$. Substitution into (13) yields $D(t) \geq b(s-2)/(t-1) > 0$, as required. Undoing complementation proves the stated sign and bound. $\square$

Completion of the proof of Theorem 1. Sufficiency was proved after Lemma 2. For $k \geq 3$, convergence to $1/2$ forces $D(\lambda) = 0$ by Lemma 3. Proposition 5 excludes this unless every border is balanced.

For $k = 1$, forbidding 0 leaves only 1^n , and forbidding 1 leaves only 0^n . Neither has half frequency or a balanced whole-word border. For $k = 2$, the words 01 and 10 have only their balanced whole-word border, so sufficiency applies. For 00 and 11, all border imbalances have the same strict sign, and $D(t)$ has no positive zero. Lemma 3 therefore excludes convergence to $1/2$. These cases exhaust all nonempty binary words. $\square$

Remark 6. The words 01 and 10 give $P(x) = (1 - x)^2$, so repeated poles are present even in the smallest balanced cases. Equation (10) is arranged to include them. The proof also allows $s < k$: longer balanced borders contribute to C and only improve (15). The sign conclusion for $D(\lambda)$ is used as a proof lemma; we do not infer a general formula for the sign of $\rho_w - 1/2$.

5 Exact checks and assistance disclosure

The proof is deductive for arbitrary k ; no finite computation is a premise of Theorem 1. Supplementary Python computations using exact integers and SymPy 1.14.0 checked the generating-function identities for all 510 forbidden words of lengths 1 through 8, through coefficient degree $2k + 8$. A separate direct enumeration checked the full distribution of ones for forbidden lengths 1 through 5 and avoiding-word lengths 0 through 10. All 524,286 nonempty words of length at most 18 were inspected for the prefix bound (12). Sturm root counts and rational intervals certified the sign of D for 140 selected mixed-sign border signatures of lengths at most 18; this selection was not an exhaustive algebraic root check through length 18.

A separate verification implementation directly enumerated all 254 forbidden words of lengths 1 through 7, checking the identities through degree 12, and reconstructed all 140 root certificates. It also verified (14) by integer polynomial coefficients for $3 \leq s \leq 256$. The accompanying reproducibility notes identify the scripts, outputs and exact coverage. These finite checks provide error detection for the separately proved identities and inequalities.

OpenAI Codex assisted with proof development, adversarial testing, Python code, literature searches and manuscript preparation. A separate model session reconstructed the argument and wrote the additional verification implementation. Such a session is not external peer review or formal proof-assistant certification. No formal proof-assistant verification is claimed.

References

- [1] Miklós Bóna, Balázs Maga and Jacob Richey, *Letter frequency in shifts of finite type with one forbidden word*, arXiv:2606.06655, version 1 (2026). arxiv.org/abs/2606.06655.
- [2] Zijian Zeng, *The Distinct-Cycle-Length Probability is Strictly Decreasing after 30*, SSRN preprint (2026), doi:10.2139/ssrn.7380519. doi.org/10.2139/ssrn.7380519.